Name:_____ **Period:**_____
Earth and Space Science

What Makes a Place Worth Protecting?

LEARNING INTENTION Today we will be learning how Earth materials differ from place to place so we can explain that landscapes are made of matter that has been changed by energy.
SUCCESS CRITERIA ● I can describe an Earth material using observable properties. ● I can explain what makes a place nationally significant. ● I can use evidence to argue that a place is scientifically worth protecting.

PART 1: First Look

You will see four photo sets, labeled A through D. Each set shows four photographs of the same place.

Site	I notice...	I wonder...
A		
B		
C		
D		

PART 2: What Makes a National Park a National Park?

All four of those sites are national parks. There are thousands of beautiful places in the United States. Only 63 are national parks. So how does a place make the list?

THE FOUR CRITERIA FOR NATIONAL SIGNIFICANCE:

The National Park Service uses real criteria. A place must pass **all four**:

#	Criterion	Ask yourself
1	**Outstanding example**	**Is this one of the best examples of this kind of place or feature anywhere in the country?**
2	Tells a national story	Does it show something important about our nation's natural or cultural history?
3	**Outstanding for enjoyment or scientific study**	**Can people learn something here that they could not learn as well anywhere else?**
4	Integrity	Is it still real and relatively unspoiled, not damaged, rebuilt, or mined out?

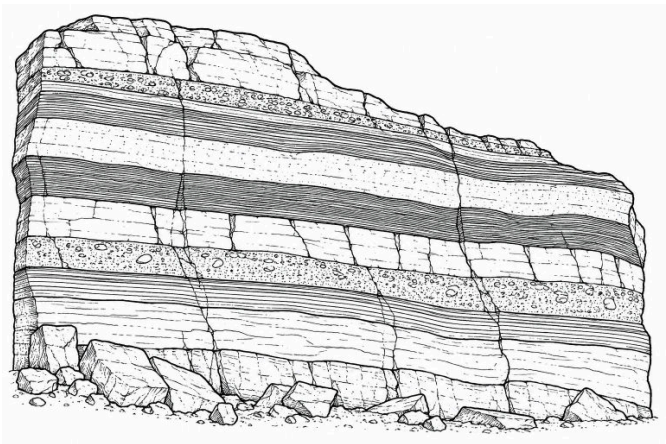
A place also has to be the right size, affordable to protect, not already protected somewhere else, and genuinely need the National Park Service specifically.

Note: the title "national park" is usually saved for the most spectacular natural areas with many different features. Smaller protected places that focus on one feature are often called national monuments instead.

At each station, you will collect evidence. You are looking for reasons these places pass Criterion 1 and Criterion 3.
There, you have three distinct jobs:

- **Photo: What does the overall landscape look like?**
- **Rock sample: What properties can I actually observe?**
- **Site Information: Why are those observable features important or unusual?**

In Part 3, you will be asked to look for layers or bands. This is what you are looking for:



Layers: Separate sections of rock that formed one on top of another over time. Each layer may have different materials, grain sizes, colors, or textures.

Bands: Stripes or zones within a rock where minerals are arranged differently.

Easy distinction:
Layers = material added over time
Bands = minerals rearranged within rock

PART 3: Station Rotation

You will now rotate between Stations A, B, C, and D. Each station has the same site photos and one or two rock samples from that park.
Rule: do not name the rock. Describe only what you can observe. If a station has two samples, circle all that apply.

Station	Color(s)	Grain or crystal size	Layers or bands?	Anything else you notice	Which criterion does your evidence support?
A		None Small Large	Yes No		☐ Criterion 1: Outstanding Example ☐ Criterion 3: Scientific Study ☐ Both Evidence:
B		None Small Large	Yes No		☐ Criterion 1: Outstanding Example ☐ Criterion 3: Scientific Study ☐ Both Evidence:
C		None Small Large	Yes No		☐ Criterion 1: Outstanding Example ☐ Criterion 3: Scientific Study ☐ Both Evidence:
D		None Small Large	Yes No		☐ Criterion 1: Outstanding Example ☐ Criterion 3: Scientific Study ☐ Both Evidence:

STATION A – Yosemite

The exposed rock here forms some of the largest cliffs of its kind in the country. Large areas of bare rock allow its crystals and other features to be clearly observed and compared. The same type of rock occurs in other places, but few expose it on such a large, continuous scale.

You are looking for reasons these places pass Criterion 1 and Criterion 3.

Look at:

- **Photo:** What does the overall landscape look like?
- **Rock sample:** What properties can I actually observe?
- **Site Information:** Why are those observable features important or unusual?

#	CRITERION	ASK YOURSELF	Which criterion does the evidence support at this station?
1	**Outstanding example**	Is this one of the best examples of this kind of place or feature anywhere in the country?	
3	**Outstanding for enjoyment or scientific study**	Can people learn something here that they could not learn as well anywhere else?	

STATION B – Hawai‘i Volcanoes

This is one of the few places in the country where new rock is still forming at Earth's surface. Scientists can observe fresh material and changes to the landscape as they happen. Older and newer surfaces can be found close together, allowing changes over time to be compared.

You are looking for reasons these places pass Criterion 1 and Criterion 3.

Look at:

- **Photo:** What does the overall landscape look like?
- **Rock sample:** What properties can I actually observe?
- **Site Information:** Why are those observable features important or unusual?

#	CRITERION	ASK YOURSELF	Which criterion does the evidence support at this station?
1	**Outstanding example**	Is this one of the best examples of this kind of place or feature anywhere in the country?	
3	**Outstanding for enjoyment or scientific study**	Can people learn something here that they could not learn as well anywhere else?	

STATION C – Badlands

This area contains an unusually large exposure of stacked rock layers. Because so many layers are visible together, scientists can compare them and study a long sequence of changes in the landscape. Different layers can be followed across large sections of the exposed landscape.

You are looking for reasons these places pass Criterion 1 and Criterion 3.

Look at:

- **Photo:** What does the overall landscape look like?
- **Rock sample:** What properties can I actually observe?
- **Site Information:** Why are those observable features important or unusual?

#	CRITERION	ASK YOURSELF	Which criterion does the evidence support at this station?
1	**Outstanding example**	Is this one of the best examples of this kind of place or feature anywhere in the country?	
3	**Outstanding for enjoyment or scientific study**	Can people learn something here that they could not learn as well anywhere else?	

STATION D – Great Smoky Mountains

This area contains extensive exposures of rock that has been changed from its original form. Scientists can examine these materials to find evidence that the rock was altered after it first formed. Similar changes can be observed in many locations across this large mountain landscape.

You are looking for reasons these places pass Criterion 1 and Criterion 3.

Look at:

- **Photo:** What does the overall landscape look like?
- **Rock sample:** What properties can I actually observe?
- **Site Information:** Why are those observable features important or unusual?

#	CRITERION	ASK YOURSELF	Which criterion does the evidence support at this station?
1	**Outstanding example**	Is this one of the best examples of this kind of place or feature anywhere in the country?	
3	**Outstanding for enjoyment or scientific study**	Can people learn something here that they could not learn as well anywhere else?	

PART 4: Looking for Patterns

1. Which site(s) (A,B,C,and/or D) had crystals or grains you could see without a hand lens? _____

2. Which site(s) (A,B,C,and/or D) had layers or bands? _____

3. One site (A,B,C,or D) cooled from something extremely hot. Which one, and what is your evidence?

Site Letter:_____

Evidence:_____

_____

PART 5: Your Argument

Pick **one** site. Using your own observations from today, argue that this place passes **Criterion 1** and **Criterion 3**.

Site I am arguing for: _____

Claim (one sentence: this place is worth protecting because...):

Evidence (specific things you observed at that site, and at one other site for comparison):

Reasoning (why that evidence shows the place passes Criterion 1 and Criterion 3):

EXIT TICKET

Complete the sentence. You will add to this idea for the rest of the unit. There is no wrong answer today.

Earth's landscapes are made of _____ that has been changed by _____.